LONGi SOLAR

# Hi-MO X10 Scientist

## LR7-60HVD

## 530~555M

More flexible installation methods, suitable for short frame clamps mounting with high mechanical loading

High efficiency with better energy generation performance

N-type TaiRay wafer & HPBC 2.0 innovative technology enhances high product reliability

HPBC 2.0 | TaiRay | N-type

Bifacial Power Generation

15Y Product Warranty | 30Y Performance Warranty

### Complete System and Product Certifications

IEC 61215,IEC 61730

IEC62941:Guideline for module design qualification and type approval

ISO9001:Quality Management System

ISO14001:Environment Management System

ISO45001:Occupational Health and Safety

TÜVRheinland CERTIFIED | TÜV SÜD | CE | TIER 1 Bloomberg NEW ENERGY FINANCE

THE smarter E AWARD



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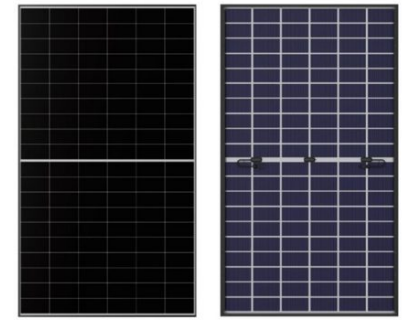
# LR7-60HVD 530~555M

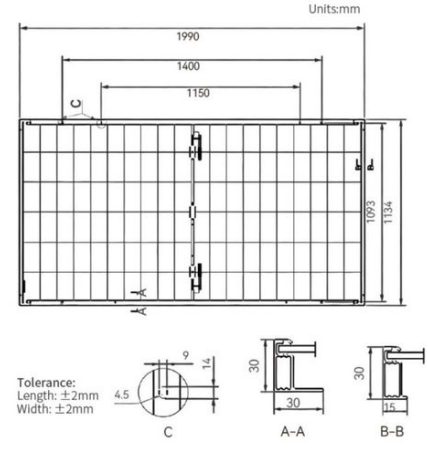
| 24.59% | 1% | 0.35% | -0.26%/°C |
|---|---|---|---|
| MAX MODULE EFFICIENCY | FIRST YEAR POWER DEGRADATION | YEAR 2-30 POWER DEGRADATION | BETTER TEMPERATUR COEFFICIENT |

## Mechanical Parameters

| | |
|---|---|
| Cell Orientation | 120 (6×20) |
| Junction Box | IP68, three diodes |
| Output Cable | 4mm²; +400/-200mm/±1400mm length can be customized |
| Glass | Dual glass, 2.0+2.0mm semi-tempered glass |
| Frame | Anodized aluminum alloy frame |
| Weight | 28 kg |
| Dimension | 1990×1134×30mm |
| Packaging | 36pcs per pallet / 180pcs per 20'GP / 792pcs per 40'HC |

Remark: 17.5m trucks are rated at 30t load. Actual loading quantity is subject to vehicle specs.






## Electrical characteristics with different rear side power gain

| Pmax/W | Voc/V | Isc/A | Vmp/V | Imp/A | Pmax gain |
|---|---|---|---|---|---|
| 535 | 45.05 | 15.00 | 37.27 | 14.36 | 0% |
| 562 | 45.05 | 15.75 | 37.27 | 15.07 | 5% |
| 589 | 45.05 | 16.50 | 37.27 | 15.79 | 10% |
| 615 | 45.15 | 17.25 | 37.37 | 16.47 | 15% |
| 642 | 45.15 | 18.00 | 37.37 | 17.18 | 20% |
| 669 | 45.15 | 18.75 | 37.37 | 17.90 | 25% |

## Electrical Characteristics

·STC: Irradiance1000W/m², Cell Temperature 25℃, AM1.5 (Test uncertainty for Pmax: ±3%)

| Module Type | LR7-60HVD-530M | LR7-60HVD-535M | LR7-60HVD-540M | LR7-60HVD-545M | LR7-60HVD-550M | LR7-60HVD-555M |
|---|---|---|---|---|---|---|
| Testing Condition | STC | STC | STC | STC | STC | STC |
| Maximum Power (Pmax/W) | 530 | 535 | 540 | 545 | 550 | 555 |
| Open Circuit Voltage (Voc/V) | 44.95 | 45.05 | 45.15 | 45.25 | 45.35 | 45.45 |
| Short Circuit Current (Isc/A) | 14.90 | 15.00 | 15.10 | 15.20 | 15.30 | 15.40 |
| Voltage at Maximum Power (Vmp/V) | 37.17 | 37.27 | 37.37 | 37.47 | 37.57 | 37.67 |
| Current at Maximum Power (Imp/A) | 14.26 | 14.36 | 14.45 | 14.55 | 14.64 | 14.73 |
| Module Efficiency (%) | 23.49 | 23.71 | 23.93 | 24.15 | 24.37 | 24.59 |

## Operating Parameters

| | |
|---|---|
| Module [T98] Max | 70℃ |
| Operational Temperature | -40℃ ~ +85℃ |
| Power Output Tolerance | 0~3% |
| Maximum System Voltage | DC1500V (IEC) |
| Maximum Series Fuse Rating | 30A |
| Nominal Operating Cell Temperature | 45±2℃ |
| Protection Class | Class II |
| Bifaciality | φPmax: 70±5% |
| Fire Rating | IEC Class C |

## Mechanical Loading

| | |
|---|---|
| Min Designed Mechanical Load | 1600Pa/1600Pa |
| Front Side Maximum Static Loading | 5400Pa |
| Rear Side Maximum Static Loading | 2400Pa |
| Hailstone Test | 25mm Hailstone at the speed of 23m/s |

## Temperature Ratings (STC)

| | |
|---|---|
| Temperature Coefficient of Isc | +0.050%/℃ |
| Temperature Coefficient of Voc | -0.200%/℃ |
| Temperature Coefficient of Pmax | -0.260%/℃ |

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